

Elicit vs. Teach, Mechanistically: Results on the TinyStories-1B Twins

MARS V — Mechanistic Understanding of Elicitation vs. Teaching

Working results document, September 2026

Setup

The experiment. Two identical copies (“twins”) of a 1B-parameter language model pretrained only on children’s stories — no arithmetic anywhere in the corpus — are fine-tuned on the same target task: natural-language addition and subtraction of 1–4-digit numbers, **What is the difference between 8 and 5784? $\rightarrow$ -5776.** The training protocol is identical for both. The *only* difference is the checkpoint the fine-tune starts from:

- **Teach twin:** the blank base model. The capability is absent.
- **Elicit twin** (“TS1B-latent”): the blank base plus two installations, neither of which shows the target task: (i) arithmetic in symbols only ($23 + 45 = 68$, 1M examples, full fine-tuning) and (ii) a word $\leftrightarrow$ symbol binding dose that only *rewrites* questions (**What is the sum of 621 and 5068? $\leftrightarrow$ 621 + 5068**) and never shows an answer. This twin can compute in symbols and knows what the words mean, but on the target task it scores 0.000: asked a question, it rewrites it instead of answering. The capability is latent.

Every construction step has a control (Table 1); the two twins are behaviourally identical on the target task before fine-tuning.

Construction step / check	value	control
symbol install: exact match on $23 + 45 =$	0.67–0.73	blank twin: 0.00
corpus census: “sum” in 439M words of stories	$21\times$	“equals” 17, “minus” 57
binding dose: question $\rightarrow$ symbols rewrite exact	0.484	blank + same dose: 0.000
sum-only dose on <i>difference</i> questions	writes “+”	binding is per word
exact match on the target task, before fine-tuning	0.000	= blank twin

Table 1: The elicit twin before fine-tuning: can compute in symbols, understands the words, still answers nothing. Everything below is about what differs inside the two twins and what that difference does.

Endpoints. Seven fine-tuned models enter the results (Table 2). With the paper’s adapter protocol (LoRA), the teach twin stops by the protocol’s own convergence rule far below the elicit twin — 0.093 with 1M examples and 0.139 with four times as many unique examples — which is where the paper’s own Figure 2 puts this model (1–2 nats per token at 1M; “stalls at lower accuracy”). Teaching this base needs full fine-tuning. Two fully fine-tuned teach endpoints are carried through every result: the *format-installed* twin (the paper’s “pre-teach format” control: the blank base first fine-tuned on a random-label dose in the target’s own format, so it already emits a number after every question; exact match 0.000) fine-tuned to convergence, which reaches 0.424; and the

blank twin fine-tuned to convergence, which reaches 0.819. The elicit twin was fully fine-tuned with the identical recipe (0.986). The format-installed pair is the design-matched comparison — both parents are the blank base plus a full-FT dose, both already output numbers, only one has the algorithm — and the blank-base pair is the better-performing one; the two taught endpoints agree on every conclusion below. LoRA endpoints are kept as robustness rows; two instruments need adapter snapshots (formation curve, weight travel) and exist only for the LoRA runs.

endpoint	parent	method	data	steps (passes)	stop	exact match
elicit LoRA	TS1B-latent	LoRA $r=512$	1M	11,500 (1.5)	converged	0.981
teach LoRA 1M	blank	LoRA $r=512$	1M	25,000 (3.2)	converged	0.093
teach LoRA 4M	blank	LoRA $r=512$	4M	39,500 (1.3)	converged	0.139
teach FT	blank	full FT	4M	62,500 (2.0)	ceiling	0.771
teach FT, converged	blank	full FT	4M	72,000 (2.3)	converged	0.819
teach FT, format-installed	format-installed	full FT	4M	58,500 (1.9)	converged	0.424
elicit FT	TS1B-latent	full FT	4M	21,500 (0.7)	converged	0.986

Table 2: Fine-tuned endpoints. LoRA learning rate 3.53×10^{-4} , full FT 2×10^{-5} ; batch 128; seed 316; training stops when validation loss stops improving (eps/k rule). Exact match = 0-shot accuracy on 1,024 held-out problems. The 4M file is the 1M file plus 3M new unique problems, disjoint from every evaluation set. “Converged” teach FT continues the 2-pass run from its checkpoint until the rule fires (optimizer state restarted).

Vocabulary used below.

- *Node*: one of 528 components of the model — 512 attention heads (32 per layer $\times$ 16 layers) and 16 MLP blocks (one per layer). Attention heads move information between token positions; MLP blocks transform it in place.
- *Circuit*: the 32 nodes that matter most for the answer. Found by attribution patching: for each node, how much the answer’s logit changes when that node’s activation is replaced by its value on a different problem (gradient $\times$ activation difference), over 128–256 problem pairs.
- *Jaccard@32*: overlap of two 32-node circuits, shared $\div$ union. Two random circuits overlap at 0.031. The *ceiling* is the overlap of two circuits of the *same* model computed from two halves of the data — how repeatable a circuit is (0.42–0.68 here). An overlap at the ceiling means “the same circuit”.
- *Answer position*: the last prompt token, where the model predicts the first token of the answer. All activation measurements are taken there unless stated.
- *Residual stream*: the model’s running internal state at a token position, a 2048-dimensional vector that every layer reads and adds to.

Reading guide. Four questions, in order: does the *behaviour* of learning differ (R1); is the resulting *machinery* the same (R2–R4); how does that machinery *come into being* during training (R5–R8); what does the latent state *mean and permit* before any target training (R9–R11). Section E puts the matched full-FT pair side by side.

A. Behaviour: the learning-curve signature

R1 Eliciting gets cheaper from the first example; teaching gets more expensive before it gets cheaper.

Metric. Prequential excess description length (EDL): as training proceeds, the extra nats the model needs to predict each answer token the first time it sees it, summed and divided

by the number of answer tokens. It is the paper’s learning-cost measure: falling EDL/token with dataset size n means each new example costs less than the last (elicitation); rising means examples are getting more informative because structure is being built (teaching). Measured on a sweep $n = 100 \dots 10^6$ with identical protocol, plus 0-shot exact match on 1,024 held-out problems.

Result. Figure 1, Table 3. Elicit twin: EDL/token strictly monotone decreasing, $4.53 \rightarrow 0.092$; exact match 0.54 already at $n=3,162$, 0.98 at 1M. Teach twin (blank base): down-up-down, minimum 0.78 at $n=14.7\text{K}$, hump peaking at 2.02 at $n=215\text{K}$, 1.40 at 1M; exact match 0.093 at 1M and 0.139 at 4M. The design-matched teach twin (format-installed base, the paper’s pre-teach-format parent) keeps the hump with the format cost removed: minimum 0.64 at 10K, peak 1.33 at 100K, 0.88 at 1M, exact match 0.079. The dataset size at which exact match crosses 0.5 differs by more than $300\times$. Under full fine-tuning, without a sweep: the elicit twin converges in 21,500 steps (0.7 of a pass, test loss 0.015, exact match 0.986); the blank teach twin runs to the two-pass ceiling (62,500 steps, 0.771) and converges 9,500 steps later at 0.819 (test loss 0.154); the format-installed teach twin converges after 58,500 steps at 0.424 (test loss 0.639).

Elicit vs. teach. Eliciting spends every example connecting a capability that already exists, so each example costs less than the last from the very first; teaching has to build the capability, so the cost rises mid-run and the model converges lower — 0.82 from the blank base, 0.42 from the format-installed base — after three times the steps.

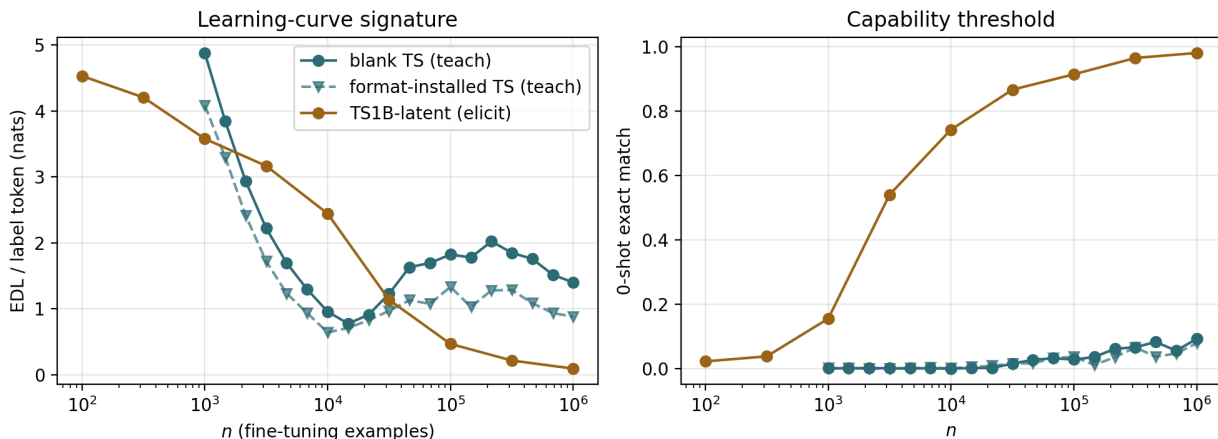


Figure 1: EDL/token (left) and 0-shot exact match (right) vs n ; teal = teach twin from the blank base, dashed teal = teach twin from the format-installed base, gold = elicit twin.

B. Machinery: is the circuit the same?

R2 The elicited child runs the circuit its parent already had; the taught child has nothing upstream to run.

Metric. Jaccard@32 between the circuit of the parent on the symbol task ($23 + 45 =$) and the circuit of the fine-tuned child on the natural-language task, read against chance (0.031) and the ceiling (0.684 for the elicited child). A parent that cannot do the task gives a noise map, which the tool flags (answer logit-diff ≈ 0).

n	elicit (TS1B-latent)		teach (blank)		teach (format-installed)	
	EDL/tok	EM	EDL/tok	EM	EDL/tok	EM
100	4.533	0.023	6.511	0.007	–	–
316	4.208	0.038	6.387	0.033	–	–
1,000	3.579	0.155	4.882	0.001	4.087	0.001
3,162	3.168	0.541	2.221	0.001	1.720	0.000
10,000	2.444	0.743	0.956	0.000	<i>0.641</i> (min)	0.001
14,678	–	–	<i>0.775</i> (min)	0.002	0.708	0.005
31,623	1.137	0.867	1.228	0.016	0.968	0.016
100,000	0.467	0.915	1.824	0.028	<i>1.333</i> (peak)	0.037
215,443	–	–	<i>2.022</i> (peak)	0.062	1.279	0.035
316,228	0.215	0.966	1.849	0.066	1.285	0.065
1,000,000	0.092	0.981	1.399	0.093	0.880	0.079
4,000,000	–	–	0.930	0.139	–	–

Table 3: Per-rung values, LoRA protocol (EDL in nats per answer token; EM = exact match; 19 rungs per teach arm, all converged; rungs shared with the elicit sweep shown plus each arm’s minimum and peak).

Result. Table 4. Parent’s symbol circuit $\leftrightarrow$ elicited child’s NL circuit: 0.455–0.524 (about 70% of the ceiling), and the top four nodes are the same four in the same order (mlp:15,14,13,12). The shared backbone is already complete at $n=3K$ (child@3K $\leftrightarrow$ child@1M: 0.391). The elicited circuit is the same under both fine-tuning methods: full-FT elicited child $\leftrightarrow$ LoRA elicited child 0.641, at the ceiling. Blank parent: no measurable circuit (logit-diff -0.33 even with 16 examples in the prompt), so the taught child has nothing to reuse.

Elicit vs. teach. Elicitation re-weights a circuit that was already there; teaching constructs machinery where there was none.

Comparison	Jaccard@32	chance	ceiling(s)
parent’s symbol circuit $\leftrightarrow$ elicited child (two eval surfaces)	0.455–0.488 / 0.524	0.031	0.684
elicited child @3K $\leftrightarrow$ @1M	0.391	0.031	0.684
elicited child, full FT $\leftrightarrow$ elicited child, LoRA	0.641 (Spearman 0.54)	0.031	0.561 / 0.684
blank parent $\leftrightarrow$ taught child	undefined (parent map is noise)	–	–
elicited FT $\leftrightarrow$ taught FT, format-installed	0.306 (Spearman 0.03)	0.031	0.561 / 0.524
elicited FT $\leftrightarrow$ taught FT, blank, 2 passes / converged	0.333 (Spearman 0.16) / 0.333 (0.28)	0.031	0.561 / 0.422
taught FT, converged $\leftrightarrow$ taught FT before continuation	0.600 (Spearman 0.57)	0.031	0.422 / 0.422
taught FT, format-installed $\leftrightarrow$ taught FT, blank, converged	0.362 (Spearman 0.18)	0.031	0.524 / 0.422
elicited $\leftrightarrow$ taught, LoRA pair (4M)	0.391 (Spearman 0.28)	0.031	0.684 / 0.524
elicited $\leftrightarrow$ taught, LoRA pair (1M)	0.231 (Spearman -0.11)	0.031	0.684 / 0.561
taught, full FT $\leftrightarrow$ taught, LoRA 4M	0.391 (Spearman 0.11)	0.031	0.422 / 0.524

Table 4: Circuit overlaps. Spearman = rank correlation of the node scores over the union of the two circuits. All maps pass the performing-regime check.

R3 The heads that read the two numbers stay the same under elicitation; teaching has to find them again.

Metric. Desiderata Component Masking (Prakash et al. 2024). For one piece of information at a time — “which number is the first operand”, “which is the second”, “which operation” — take a problem and a counterfactual problem that differs only in that piece, and learn the smallest set of attention heads such that copying those heads’ activations from the counterfactual run

into the clean run makes the model answer the counterfactual problem. That set is the heads carrying that piece of information (its “role”). The set is validated by its counterfactual-flip accuracy against the ceiling obtained by copying every head. Role sets are compared across models by Jaccard (chance ≈ 0.05).

Result. Table 5. Parent $\leftrightarrow$ LoRA-elicited child, flip accuracy at the ceiling for every set: first operand 0.879 (29 of 33 heads shared), second operand 0.900, operation 0.750. The operand heads are almost all of layer-0 attention (30 of 32 heads). The full-FT elicited child keeps the same heads (0.811 / 0.833 vs the LoRA child, 0.694 / 0.857 vs the parent). The full-FT taught child (exact match 0.77) also has compact operand sets at the ceiling; they overlap the elicited child’s at 0.587 / 0.763, the parent’s at 0.763 / 0.684, and the full-FT elicited child’s at 0.700 / 0.632 — 28 of its 31 first-operand heads are in the taught set, which adds nine more. The format-installed taught child (0.42) has compact sets at its ceilings too: 0.879 / 0.659 vs the parent’s engine, 0.659 / 0.690 vs the elicited child; the converged blank-base one (0.82): 0.763 / 0.658 and 0.587 / 0.737. The two LoRA-taught children (exact match 0.09 / 0.14) have no compact set at all: best flip accuracy 0.008–0.04 against ceilings of 0.14–0.20.

Elicit vs. teach. Elicitation keeps the parent’s operand-reading heads nearly intact (0.88–0.90 overlap); teaching has to rediscover them and lands on a larger set that contains most of the same heads (0.6–0.76) — and a taught model that never learned the task has no such heads at all.

Comparison (first operand / second operand)	$ A $	$ B $	Jaccard	flip acc. / ceiling
parent $\leftrightarrow$ elicited (LoRA), symbol surface	30 / 27	32 / 30	0.879 / 0.900	at ceiling
parent $\leftrightarrow$ elicited (LoRA), operation	25	24	0.750	at ceiling
elicited (full FT) $\leftrightarrow$ elicited (LoRA)	31 / 25	36 / 30	0.811 / 0.833	0.98 / 0.98
elicited (full FT) $\leftrightarrow$ parent	31 / 25	30 / 27	0.694 / 0.857	–
taught (full FT) $\leftrightarrow$ elicited (LoRA)	37 / 37	36 / 30	0.587 / 0.763	0.81 / 0.79 (0.81 / 0.80)
taught (full FT) $\leftrightarrow$ elicited (full FT)	37 / 37	31 / 25	0.700 / 0.632	–
taught (full FT) $\leftrightarrow$ parent	37 / 37	30 / 27	0.763 / 0.684	–
taught (FT, format-installed) $\leftrightarrow$ parent	32 / 41	30 / 27	0.879 / 0.659	0.59 / 0.54 (0.59 / 0.55)
taught (FT, format-installed) $\leftrightarrow$ elicited (LoRA)	32 / 41	36 / 30	0.659 / 0.690	–
taught (FT, blank, converged) $\leftrightarrow$ parent	37 / 36	30 / 27	0.763 / 0.658	0.85 / 0.81 (0.85 / 0.83)
taught (FT, blank, converged) $\leftrightarrow$ elicited (LoRA)	37 / 36	36 / 30	0.587 / 0.737	–
taught (LoRA 4M) $\leftrightarrow$ elicited (LoRA)	12 / 21	36 / 30	0.043 / 0.041	0.01 / 0.03 (0.20 / 0.14)
taught (LoRA 1M) $\leftrightarrow$ elicited (LoRA)	9 / 21	36 / 30	0.125 / 0.085	0.02 / 0.04 (0.15)

Table 5: Head roles by Desiderata Component Masking (heads only, 64–128 counterfactual pairs). $|A|$, $|B|$ = size of the two role sets; flip accuracy is reported for set A .

R4 The two elicited children run one circuit; every taught child runs a different one.

Metric. Jaccard@32 and score rank-correlation between the circuits of two fine-tuned children, each read against the two circuits’ ceilings; the layer histogram of each circuit’s top-16 nodes. For the LoRA-1M pair also the change at the level of connections between nodes (edge-level $\Delta S = 1 - \text{Jaccard}$ over the 256 strongest node→block edges).

Result. Table 4, Figure 2. The two elicited children — one LoRA, one full fine-tune — overlap at 0.641, at the ceiling. Every taught child sits apart from them: 0.306 for the format-installed full-FT pair, 0.333 for the blank-base full-FT pair (before and after convergence), 0.391 for the LoRA-4M pair, 0.231 for the LoRA-1M pair (about 60% of the ceiling), with

score correlations between -0.11 and 0.28 . Taught children agree with each other at 0.36 – 0.60 . What all circuits share is the late-layer MLP engine (`mlp:15,14,13,7`); what differs is the rest: taught circuits add `mlp:0,10,11` (the format-installed one also `mlp:1,3,4,5,6`) and their own attention heads (eight layer-0 heads over the raw digits under LoRA; heads in layers 4–7 and 15 under full FT), elicited circuits use `mlp:8,9` and three layer-7 heads. LoRA-1M pair: $\Delta S_{\text{node}} = 0.769$, $\Delta S_{\text{edge}} = 0.739$, both far above the noise floors ($0.32 / 0.61$).

Elicit vs. teach. Elicitation lands on one circuit whichever way it is trained; teaching builds a related but different one — it reconverges on the same late-layer MLPs but fetches and routes the operands through attention heads of its own.

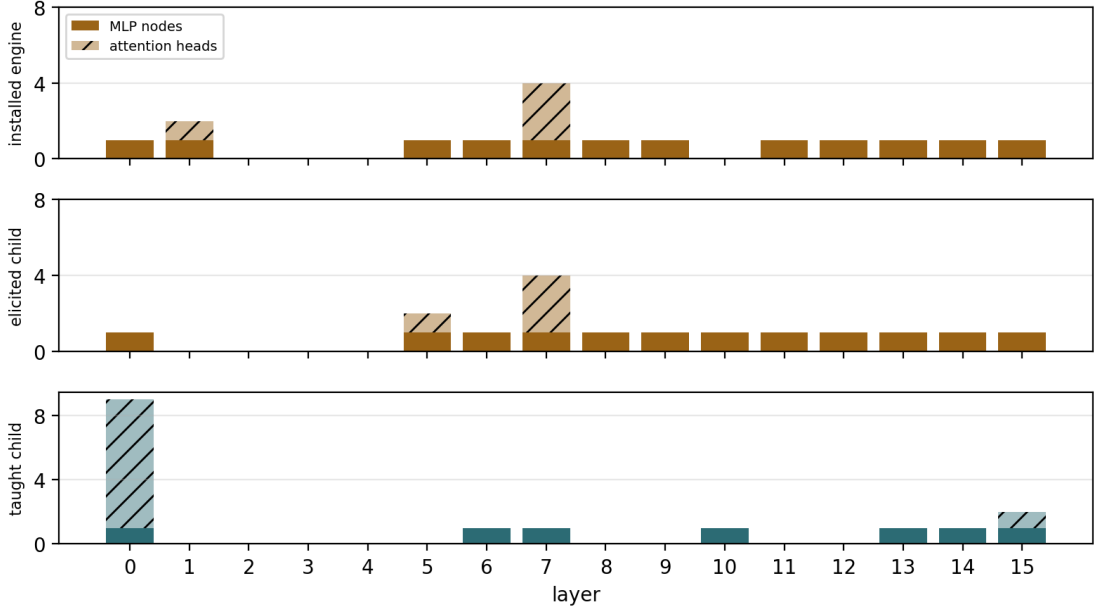


Figure 2: Top-16 node layer histograms (solid = MLP, hatched = attention) for the parent’s symbol circuit, the LoRA-elicited child and the LoRA-1M taught child.

C. Dynamics: how the machinery comes into being

R5 The elicited circuit exists at step 1; the taught circuit is assembled mid-run, during the EDL hump.

Metric. Adapter snapshots were saved along the LoRA runs. For each snapshot, compute the circuit and its Jaccard@32 with the run’s final circuit; read against chance (0.031) and the ceiling.

Result. Figure 3. Elicit: 0.488 at step 1 ($16\times$ chance), a brief dip to ~ 0.42 while the input interface is wired (step 154), then locked from about step 1.5K of 11.5K (0.561 ; 82% of the ceiling, 97% of the J@64 ceiling by 8.4K). Teach (1M): at the noise floor (0.22 – 0.28) until about step 1.3K, then $0.27 \rightarrow 0.44$ over steps 1.3K–5.4K — the region where its EDL peaks.

Elicit vs. teach. The elicited circuit is present before training starts and stops changing by one-eighth of the run; the taught circuit does not exist until a thousand steps in and is built exactly while the learning cost is rising.

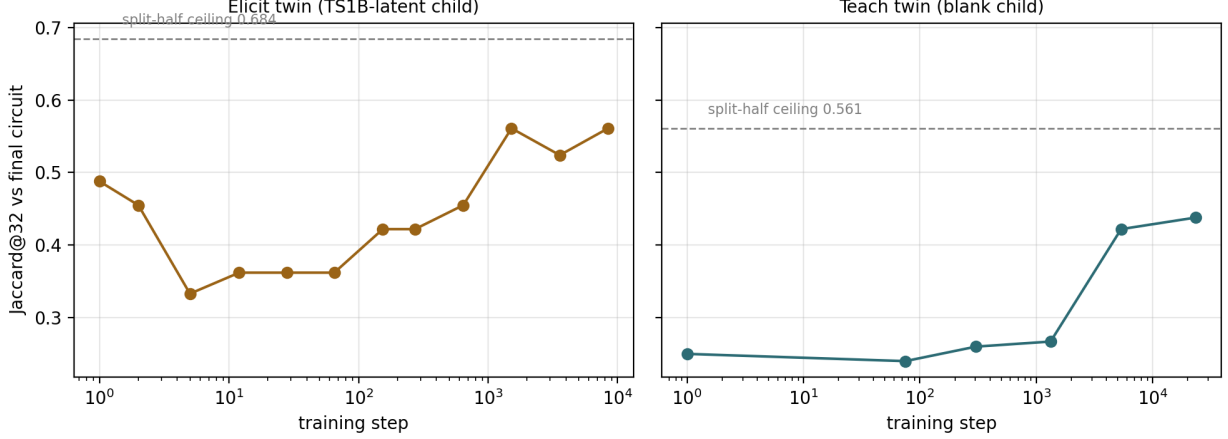


Figure 3: Circuit formation, elicit (left) vs teach (right); dashed = split-half ceiling.

R6 Elicitation’s gradient dies out; teaching’s keeps growing.

Metric. The global gradient norm before clipping, logged at every update: its peak, its mean over the first 1%, the middle and the last 10% of steps, and the cumulative gradient mass $\sum_t \|g_t\|$. For the LoRA runs also how that mass splits across attention query/key weights (QK), attention value/output weights (VO) and MLP weights; the full-FT trainer logs the global norm only. Norms are comparable within a method, not across methods (different parameter sets).

Result. Table 6, Figure 4. Both arms start at the same scale (first-1% mean 0.18 vs 0.17). Elicit decays $3.3\times$ to a floor of 0.056 (peak 0.44 at step 96). Teach grows $40\text{--}50\times$: last-10% mean 6.8 at 1M and 7.8 at 4M, peaks of 17.4 and 22.0 near the end of training. Cumulative mass 783 (elicit) vs 116,877 (teach 1M) and 220,685 (teach 4M): $150\text{--}280\times$. Across the sweep the arms are indistinguishable up to $n=10\text{K}$ and diverge from 31.6K ($7\times \rightarrow 55\times \rightarrow 95\times \rightarrow 150\times$). Teach’s gradient mass moves out of the MLPs (share $0.47 \rightarrow 0.01$) into attention (VO share 0.87 at 1M, 0.93 at 4M). Full-FT pair: elicit peaks at step 1 (34.0) and decays $3.4\times$ to a floor of 1.4; teach starts lower (2.3), rises to its peak at step 3,844 (32.0), holds 5.3 mid-run and ends at $3.4 - 1.5\times$ its starting level; cumulative mass 44,021 vs 353,277 ($8\times$; $2.8\times$ per step over $2.9\times$ the steps); the continuation to convergence adds 9,500 steps at a flat norm (first-to-last ratio 1.1, 30,954 more mass). From the format-installed parent: 2.69 at the start, peak 70.9 at step 50,860 (the largest single update anywhere), 4.86 mid-run, 3.77 at the end ($1.4\times$ its start); mass 378,170 (6.46 per step, $8.6\times$ elicit).

Elicit vs. teach. Elicitation converges and its gradient dies (decaying $3.3\text{--}3.4\times$ under either method); teaching generates gradient pressure that grows with the structure being built — $40\text{--}50\times$ under LoRA, $1.5\times$ under full FT — accumulating $8\times$ to $280\times$ the gradient mass, and under LoRA aimed at attention weights, the operand-fetching heads of R4 being written.

R7 Teaching writes twice as much into the weights and buys forty times less per unit written.

Metric. Relative weight change $\|\Delta W\|/\|W\|$ between child and parent (exact from the LoRA factors, or from the checkpoint difference for full fine-tuning); total travel $\|\Delta W_t\|_F$ along the

	elicit LoRA 1M	teach LoRA 1M	teach LoRA 4M	elicit FT	teach FT, blank (+ cont.)	teach FT, format-inst.
steps	11,500	25,000	39,500	21,500	62,500 (+ 9,500)	58,500
peak gradient norm (step)	0.441 (96)	17.4 (23,657)	22.0 (36,931)	34.0 (1)	32.0 (3,844)	70.9 (50,860)
mean norm: first 1% / middle / last 10%	0.183 / 0.060 / 0.056	0.169 / 5.21 / 6.80	0.160 / 6.61 / 7.78	4.80 / 1.85 / 1.39	2.30 / 5.31 / 3.43	2.69 / 4.86 / 3.77
first 1% ÷ last 10%	×3.3 (decays)	×0.02 (grows ~40×)	×0.02 (grows ~50×)	×3.4 (decays)	×0.7 (grows 1.5×)	×0.7 (grows 1.4×)
cumulative gradient mass $\sum_t \ g_t\ $	783	116,877	220,685	44,021	353,277 (+ 30,954)	378,170
gradient mass per step	0.068	4.68	5.59	2.05	5.65 (3.26)	6.46
mass share QK / VO / MLP	.18 / .55 / .27	.12 / .87 / .01	.05 / .93 / .01	—	—	—

Table 6: Raw gradient strength (pre-clip global norm at every update). LoRA and full-FT norms are on different parameter sets and are compared within a method only.

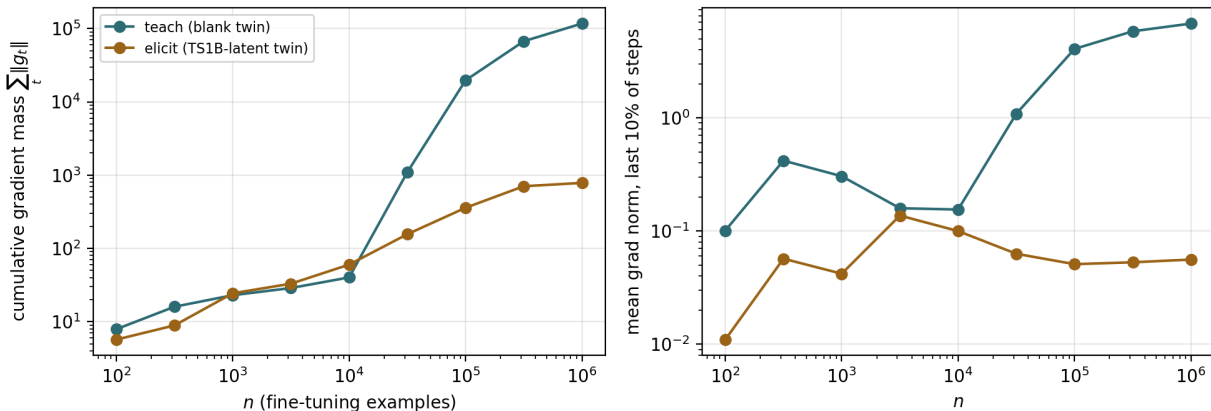


Figure 4: Cumulative gradient mass (left) and late-run gradient norm (right) vs n ; teal = teach, gold = elicit.

LoRA runs from adapter snapshots; capability per unit travel = final exact match $\div$ total travel.

Result. Table 7. LoRA: relative write 0.212 (teach 1M) and 0.264 (teach 4M) vs 0.095 (elicit), 2.2–2.8 $\times$; total travel 457 vs 120 (3.8 $\times$) over twice as many steps; capability per unit travel 2.0×10^{-4} vs 8.1×10^{-3} ($\sim 40\times$). Full FT, no adapters involved: 0.094 (teach, blank, 2 passes), 0.102 (converged), 0.086 (format-installed parent) vs 0.050 (elicit), 1.7–2.0 $\times$, with the same effective rank of the change (480–580 vs 554 of 2048). The travel *profile* is the same in both arms (a burst, then decay): the optimiser’s normalisation compresses the 150 $\times$ gradient gap of R6 into a 3.8 $\times$ step-size gap.

Elicit vs. teach. Teaching writes two to four times more into the weights than elicitation and gets forty times less capability per unit written, under both fine-tuning methods — eliciting connects what exists, teaching writes what does not.

R8 Elicitation’s change to the internal state is a different vector for every problem; teaching’s is half the same vector every time.

Metric. Run the same 256 problems through parent and child (both in fp32) and take the difference of their residual streams at the answer position, layer by layer. Its size relative to the parent’s state is reported, and its *direction*: stack the 256 difference vectors and compute the fraction of their energy in the single best direction (PC1; 1.0 means every problem’s difference is the same vector, and a set of unrelated vectors scores near the value of the parent’s own states, given for reference). The same PC1 is recomputed after projecting out the directions

	elicit (TS1B-latent)	teach (blank)
relative write $\ \Delta W\ /\ W\ $, LoRA 1M	0.095	0.212
relative write, LoRA 4M	—	0.264
relative write, full FT (4M): blank 2 passes / converged / format-installed	0.050	0.094 / 0.102 / 0.086
steps to convergence (LoRA 1M)	11,500	23,496
peak / final writing speed per step (LoRA 1M)	0.218 / 0.018	0.306 / 0.022
total weight travel $\ \Delta W\ _F$ (LoRA 1M)	120	457
final exact match (LoRA 1M)	0.981	0.093
capability per unit travel (LoRA 1M)	8.1×10^{-3}	2.0×10^{-4}

Table 7: Weight-space write, elicit vs teach (travel curves in 05_weights/fig_weight_travel).

that encode the answer token, so that “the same vector” cannot simply be the answer.

Result. Figure 5, Table 8. PC1 of the shift at the read-out: elicit 0.063 (LoRA) and 0.056 (full FT), with the parent’s own states at 0.23 — no shared direction; teach 0.528 (LoRA 4M), 0.521 / 0.526 (full FT from the blank base, 2 passes / converged) and 0.567 from the format-installed parent, and still 0.52–0.56 after the answer-token directions are removed. The format-installed parent already emits a number after every question, so its child has no output format to learn — and its shift carries the same shared component: the component is the algorithm’s mode, not the format. Cosine of each shift to the mean shift: elicit 0.26 / 0.23, teach 0.26 / 0.74 / 0.80. Size and depth depend on the fine-tuning method (LoRA-taught final-layer write 14–29 $\times$ the parent’s norm, full-FT 1.2–1.4 $\times$, elicit 1.7–1.9 $\times$), the direction does not.

Elicit vs. teach. Elicitation changes the model’s state by a problem-specific vector — the answer routed to the output — because the parent already sits in a problem-specific state; teaching’s change is dominated by one shared component in every problem, because the blank parent’s state at the answer position carries no problem information and the model has to be moved into answering mode before any content can be added.

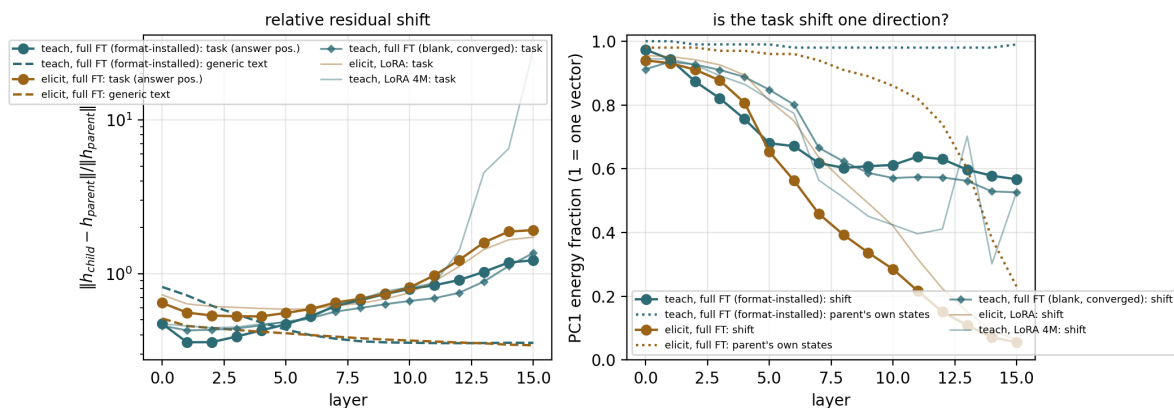


Figure 5: Matched full-FT pair (thick) with the LoRA endpoints thin. Left: relative residual shift by layer at the answer position (solid) and on generic story text (dashed). Right: PC1 energy fraction of the 256 per-problem shift vectors, with the parent’s own states dotted.

	elicit LoRA 1M	teach LoRA 4M	elicit FT	teach FT, blank (conv.)	teach FT, format-inst.
final exact match	0.981	0.139	0.986	0.819	0.424
PC1 of shift at the read-out (parent’s own states)	0.063 (0.23)	0.528 (0.98)	0.056 (0.23)	0.526 (0.98)	0.567 (0.99)
PC1 after removing answer-token directions	0.072	0.520	0.061	0.528	0.561
cosine of each shift to the mean shift	0.26	0.26	0.23	0.74	0.80
shift at answer position, final layer (relative)	1.72	28.6	1.92	1.36	1.22

Table 8: Residual-stream shift parent→child at the answer position, 256 problems, fp32. Blank-base teach FT before continuation: PC1 0.521, final-layer shift 1.42; LoRA-1M teach: PC1 0.351, shift 14.1.

D. The latent state: what it means and what it permits before training

R9 In the elicit parent, the words already reach the arithmetic machinery.

Metric. Render each problem two ways — `What is the sum of 621 and 5068?` and `621 + 5068 =` — and record every node’s activation at the answer position. Per node: the mean cosine between the two renderings of the *same* problem minus the mean cosine between renderings of *different* problems. A positive value means the words put the node into the same problem-specific state the symbols do. Compared between the symbol-only engine (no binding dose) and the elicit parent (binding dose installed). A second, geometry-only check: how problem-specific the answer-position state itself is, measured as PC1 and cosine-to-mean across problems.

Result. Symbol-only engine: MLP index ≈ 0.00 – 0.01 at every layer. Elicit parent: $+0.07$ – 0.13 across layers 7–15, about $10\times$ (attention heads are excluded: they share digit tokens across renderings). Geometry of the answer-position state at the final layer: elicit parent cosine-to-mean 0.51, PC1 0.23 (problem-specific); blank base 0.99 and 0.98, and the format-installed base 0.99 and 0.99 (the same state for every problem: emitting a number is not knowing which).

Elicit vs. teach. Before any target training, a natural-language question already activates the elicit parent’s arithmetic MLPs and leaves it in a problem-specific state; the teach parent has no arithmetic MLPs to activate and is in the same state for every problem.

R10 The elicit parent already “thinks” the answer before it can say it; the blank base does not. After training, elicited models finish the answer one layer earlier than taught ones.

Metric. Read the model’s intermediate state at each layer as if it were the output. Logit lens: apply the final unembedding to the layer- ℓ state. J-lens (Jacobian lens): first transport the state to the final-layer basis with the average Jacobian $\partial h_{\text{final}}/\partial h_\ell$ over held-out story text, then unembed — this reads what the state is *disposed to make the model say*, and it is where the answer can appear before the model emits it. R-lens: the same transport with a relevance-propagation backward pass. Scored token = the first digit chunk of the answer (for negative answers the sign is fed in first, so predicting a minus sign never counts). Read-outs per layer: the answer token’s median rank among 128K vocabulary entries (chance 64K), and the “settled depth”, the first layer from which the token stays top-ranked to the end.

Result. Figure 6, Table 9. Elicit parent on the NL task, J-space rank of the answer: 2,606 at layer 12, 510 at layer 14, 174 in the actual output, and it prefers the correct answer over a mismatched problem’s answer by 2.1 nats — while it emits a rewrite, not a number. Blank

base: 55–80K at every layer, in all three lenses, logit-diff 0.0. The format-installed base is the control that separates “a number is coming” from “this number”: it ranks the answer at 452 at the output because it emits numbers, but its logit-diff is 0.0 at every layer. After training: the LoRA-elicited child ranks the answer 139 / 6 / 1 at layers 12 / 13 / 14 and settles at layer 14 — the same layer-by-layer trajectory as the parent’s engine on symbols (122 / 6 / 1). The matched full-FT pair: elicited 338 / 8 / 1, settled at layer 14 (quartiles 14–14), top-1 0.80 at layer 14; taught 4,196 / 1,916 / 79, settled at layer 15 (quartiles 15–15), top-1 0.12 at layer 14 rising to 0.80 at layer 15; converged: 4,123 / 1,832 / 138, settled at layer 15, top-1 0.16 → 0.84; from the format-installed parent: 27,044 / 15,058 / 961, settled at layer 15, top-1 0.11 → 0.54. LoRA-taught children: 17–20K at layer 14, then 6–8 at layer 15.

Elicit vs. teach. The elicit parent carries the answer in its verbalizable space before any target training and elicitation only amplifies that read-out; the blank base carries nothing, and a taught model computes the answer one layer later than the engine even when it solves the task.

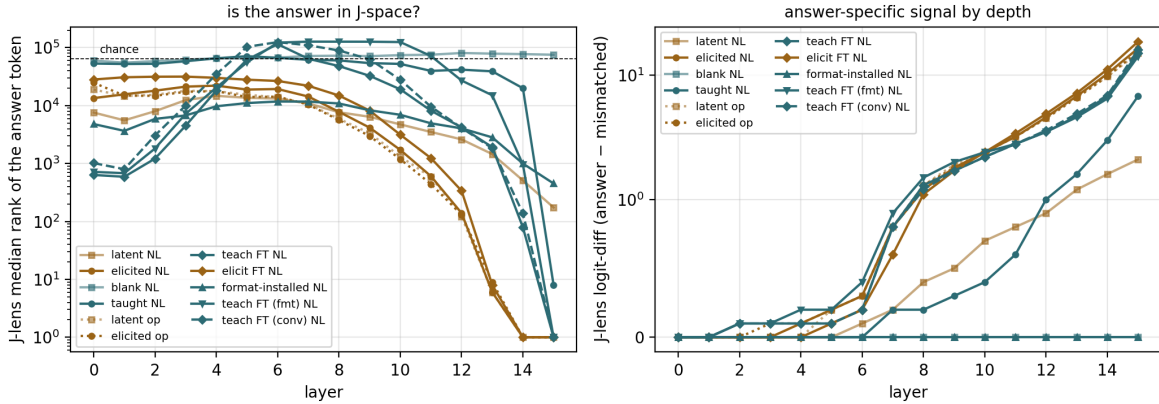


Figure 6: J-lens read-out of the answer token by layer: median rank (left, chance 64K) and logit-diff against a mismatched problem’s answer (right). Squares = parents, circles = LoRA children, diamonds = the full-FT pair; dotted = symbol surface.

model	surface	own acc	rank L12 (logit / J / R)	rank L14 (logit / J / R)	read-out	ld
elicit parent (TS1B-latent)	NL	0.047	5,704 / 2,606 / 2,730	436 / 510 / 424	174	+2.1
blank base	NL	0.000	76,049 / 80,840 / 78,243	77,707 / 77,320 / 79,116	75,441	0.0
format-installed base	NL	0.004	8,394 / 4,036 / 5,417	1,106 / 997 / 998	452	0.0
elicited child (LoRA)	NL	0.957	1,584 / 139 / 464	1 / 1 / 1	1	+16.5
elicited child (full FT)	NL	0.988	1,797 / 338 / 492	1 / 1 / 1	1	+18.7
taught child (full FT, blank, converged)	NL	0.844	18,988 / 4,123 / 9,620	170 / 138 / 459	1	+16.1
taught child (full FT, format-installed)	NL	0.535	23,909 / 27,044 / 25,529	381 / 961 / 1,848	1	+14.2
taught child (LoRA 4M)	NL	0.117	54,896 / 41,722 / 41,666	40,496 / 17,183 / 21,998	6	+7.6
taught child (LoRA 1M)	NL	0.109	51,167 / 41,677 / 44,335	35,620 / 19,933 / 21,156	8	+6.8
elicit parent (TS1B-latent)	symbols	0.836	1,224 / 122 / 334	1 / 1 / 1	1	+15.1
elicited child (LoRA)	symbols	0.855	1,360 / 131 / 399	1 / 1 / 1	1	+15.5

Table 9: Median rank of the answer’s first digit token among 128K entries under the three lenses; “own acc” = the model’s own first-digit-token accuracy on the 256 problems; read-out = the last layer, where all lenses equal the model’s output; ld = logit-diff at the read-out, the model’s preference for the correct answer over a mismatched problem’s answer (nats). Blank-base teach FT before continuation: 18,502 / 4,196 / 9,429, then 141 / 79 / 377, own acc 0.797, ld +15.2.

R11 The latent capability can be switched on with the weights frozen; the absent one cannot.

Metric. Exact match on the target task under interventions that train nothing. (a) Self-chain: let the parent rewrite the question in its own symbols, then feed it its own rewrite. (b) Donor patching: take the fine-tuned child’s activations at the 32 circuit nodes and write them into the parent at the answer position, either as one mean vector or as each problem’s own activations (“per-prompt states”); the same with 32 random nodes as control. Done for the elicit side (child into the elicit parent) and the teach side (taught child into the blank base).

Result. Figure 7, Table 10. Elicit parent: direct 0.000; self-chain **0.328** ($= 0.484 \times 0.668$, rewrite accuracy $\times$ symbol accuracy — this product held on four model variants); per-prompt states of the LoRA child 0.125, of the full-FT child 0.121 (random nodes: 0.000); the mean vector changes the output format only. Teach side: blank base plus the same doses 0.000; blank plus chain 0.000 (a parser without an engine); taught donor into the blank base ≤ 0.004 in every condition, for all four taught children; the format-installed base with its own taught child’s per-prompt states 0.016.

Elicit vs. teach. The elicit parent is one composition step or one activation patch away from the task because the machinery is there and only has to be switched on; the teach parent has nothing underneath to switch on, so no patch from any taught model helps.

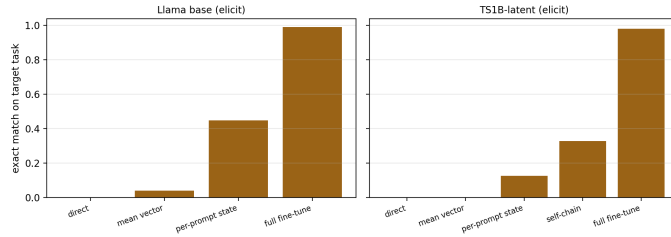


Figure 7: Intervention ladder on the elicit parent (weights frozen except the last bar).

Patient + intervention (32 circuit nodes)	condition	format	exact match
elicit parent + LoRA child’s mean vector	circuit nodes	0.164	0.000
elicit parent + LoRA child’s per-prompt states	circuit nodes	0.359	0.125
	random-32 nodes	0.051	0.000
elicit parent + full-FT child’s per-prompt states	circuit nodes	0.387	0.121
elicit parent self-chain (no donor)	own rewrite + “= ”	–	0.328
blank + binding dose, self-chain	parser without engine	–	0.000
blank + taught donor (LoRA 1M / LoRA 4M / full FT / converged)	every condition	≤ 0.02	≤ 0.004
format-installed base + its taught child’s per-prompt states	circuit nodes	0.836 (base 1.000)	0.016

Table 10: Zero-training interventions. “Format” = fraction of outputs that are a number at all.

E. The full-fine-tune endpoints at a glance

Both regimes under one method. The elicit twin and two teach twins, each fully fine-tuned with the identical recipe on the identical 4M file to convergence. The format-installed teach twin is the design-matched comparison (both parents already output numbers); the blank-base teach twin is the better-performing one. They agree on every row.

	elicit FT	teach FT, format-installed	teach FT, blank (conv.)	result
exact match (0-shot)	0.986	0.424	0.819	R1
steps to stop (passes)	21,500 (0.7)	58,500 (1.9)	72,000 (2.3)	R1
held-out test loss (nats/token)	0.015	0.639	0.154	R1
circuit overlap with the elicit-FT circuit	–	0.306	0.333	R4
circuit overlap with the LoRA-elicited child	0.641	0.333	0.333	R2, R4
top-8 nodes sufficient / necessary	0.923 / 0.943	0.983 / 0.988	0.950 / 0.965	validation
operand-reading heads vs the parent’s engine (first / second)	0.694 / 0.857	0.879 / 0.659	0.763 / 0.658	R3
gradient norm, first 1% → last 10% of steps	4.80 → 1.39 (decays 3.4×)	2.69 → 3.77 (grows 1.4×)	2.30 → 3.43 (grows 1.5×)	R6
cumulative gradient mass	44,021	378,170	353,277 (+ 30,954)	R6
relative weight write $\ \Delta W\ /\ W\ $	0.050	0.086	0.102	R7
PC1 of the state shift (after removing answer content)	0.056 (0.061)	0.567 (0.561)	0.526 (0.528)	R8
answer settled depth, median (quartiles)	layer 14 (14–14)	layer 15 (15–15)	layer 15 (15–15)	R10
J-space rank of the answer at layers 12 / 13 / 14	338 / 8 / 1	27,044 / 15,058 / 961	4,123 / 1,832 / 138	R10
child’s states patched into its own parent, per-prompt (mean)	0.121 (format only)	0.016 (0.000)	0.004 (0.000)	R11

Table 11: The full-FT endpoints, one row per instrument.

Validation (what licenses the comparisons above)

- **Every circuit is real and load-bearing.** True activation patching (copy the top nodes’ clean activations into a corrupted run, or corrupt them in a clean run): top-8 of 528 nodes recover / destroy the answer at LoRA-elicited 0.997 / 0.998, LoRA-4M taught 0.976 / 0.987, full-FT taught 0.950 / 0.966 (converged 0.950 / 0.965; format-installed 0.983 / 0.988), full-FT elicited 0.923 / 0.943; top-32 ≥ 0.996 everywhere. Every overlap above compares sets that provably carry the computation.
- **Teach-side results were re-measured on performing taught models.** The first taught endpoint (LoRA 1M, exact match 0.093) had not learned the task. Every teach-side instrument was re-run on the LoRA-4M and full-FT taught endpoints, with the full-FT elicited endpoint as the method control. Three earlier readings changed and are stated in their revised form above: taught operand roles are mostly shared rather than absent (R3), the machine difference is partial rather than total (R4), and the size of the taught state shift depends on the fine-tuning method (R8).
- **Lens depth is scored on the first digit token.** Scoring the raw first token credited a taught child with layer-5 formation on 48 problems — all subtractions with a negative answer, whose first token is the minus sign, produced by a comparison circuit. With the sign fed in first the effect vanishes; the other models were unaffected.
- **Construction controls.** The same doses on the blank twin change nothing on the target (0.000 in every probe); a sum-only binding dose leaves “difference” unbound; the composition law self-chain = rewrite $\times$ compute held on four variants.